

Analysis of Value at Risk (VaR) Methodologies and Portfolio Risk Assessment (IBIT - LVMH)

12/04/2026

Introduction - assets description

The objective of this report is to analyze the risk profile of a financial portfolio using different **Value at Risk (VaR)** methodologies, based on the results obtained in the empirical analysis.



Value at Risk is a widely used risk measure that quantifies the maximum potential loss of a portfolio over a specific time horizon and confidence level. This report focuses on five main approaches: **sigma flat**, **EWMA (Exponentially Weighted Moving Average)**, **Monte Carlo Simulation**, **Historical Var** and **Historical Simulation**. The analysis has been made on a portfolio consisting of two assets:

- IBIT (iShares Bitcoin Trust), engineered to reflect the performance of the price of bitcoin
- LVMH (Moët Hennessy Louis Vuitton), which operates as a highly diversified, physical conglomerate within the global luxury sector. Encompassing 75 prestigious

brands spanning fashion, leather goods, perfumes, cosmetics, watches, and jewelry.

Almost every VaR computed is expressed as a multiplicative factor related to the percentage of capital we are risking, i.e. if the VaR(95%) at T=100 is 0,45 it means that the Value at risk is 45% of the current portfolio value. To compute the parameters we took in consideration 6 months of past data

METHODOLOGIES

SIGMA FLAT VaR

calculated assuming a normal distribution of returns with constant volatility. It uses the formula

$VaR = Z \times \sigma \times \sqrt{T} - E \times T$, where Z is the confidence level Z-score. The factors σ , E and T represent standard deviation (volatility), mean, and time horizon, respectively.

	IBIT	LVMH
Mean	-0,40%	-0,13%
Variance	0,00107186	0,0003831
St. Dev.	0,03273921	0,01957302

- we notice that the volatility of IBIT is particularly high, as one could expect
- The **covariance** and **correlation** are respectively -1,07E-05 and -0,0167. Namely this result highlights that **the two assets do not belong to the same economic sector.**

Then the **Portfolio mean** and **Variance** were computed through the formulas:

- $E_p = (1/2 \times E_1) + (1/2 \times E_2)$
- $\sigma^2_p = (1/4 \times \sigma_1^2) + (1/4 \times \sigma_2^2) + (1/2 \times Cov_{12})$

where we used $\frac{1}{2}$ because the portfolio is assumed to be equibalanced.

At T=100, the VaR is

confidence level	IBIT	LVMH	JOINT
95%	0,938	0,449	0,575
99%	1,161	0,583	0,704
99,50%	1,243	0,632	0,751

The values 1,161 and 1,243 are **not realistic**, in the sense that we cannot lose more than what we invested. Those numbers have to be intended as 1, which is reasonable considering its recent downward trend and its extremely high volatility.

EWMA VaR

EWMA VaR is a parametric approach that assumes normal returns but dynamically updates volatility, giving more weight to recent market observations. Variance is updated via $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) R_{t-1}^2$, where λ (0, 94) is the decay factor. This makes the risk metric highly responsive to sudden market shocks and volatility clustering compared to the constant sigma approach. Also the covariance is computed similarly.

confidence level	IBIT	LVMH	JOINT
95%	0,901	0,388	0,552
99%	1,109	0,495	0,671
99,50%	1,185	0,535	0,715

Monte Carlo VaR

Monte Carlo VaR generates thousands of random, hypothetical price paths for assets.. The portfolio is reevaluated under each simulated scenario to build a comprehensive, mathematically generated distribution of potential future returns. The VaR is extracted by identifying the worst loss at the desired confidence level (percentile) directly from this generated distribution. VBA has been used to implement the method. T=100 N=10000:

confidence level	IBIT	LVMH	JOINT
95%	0,946	0,448	0,571
99%	1,151	0,585	0,706
99,50%	1,224	0,637	0,753

We notice that the Monte Carlo Method converges to the sigma flat one, as we used the same data as input. The method is nowadays used with more sophisticated sigmas.

Historical Simulation VaR

Historical Simulation VaR measures risk by applying actual past market returns to the current portfolio, assuming future movements will mirror history. It creates a series of simulated future portfolio values without assuming any specific statistical distribution. The VaR is determined by sorting these simulated outcomes from worst to best and finding the loss at the chosen percentile.

confidence level	IBIT	LVMH	JOINT
95%	0,496	0,236	0,291
99%	0,688	0,433	0,430
99,50%	1,315	0,789	0,676

Historical VaR

Historical VaR is a direct risk measurement approach that evaluates potential future losses by looking at the actual past returns of an asset or portfolio. It simply ranks these historical daily returns from worst to best and identifies the maximum expected loss at a chosen confidence level. It assumes that past market behavior is a good indicator of future risks. It shows the same results as the previous method, however, they are different. In our specific portfolio, we are analyzing IBIT (Bitcoin ETF) and LVMH, which are straightforward, linear "spot" assets. Because stocks and standard ETFs do not have expiration dates, strike prices, or complex changing sensitivities like options do, their risk profile is constant. Consequently, looking directly at their historical returns (Historical

VaR) and revaluing them under past market shocks (Historical Simulation) are functionally identical processes that will yield the exact same mathematical result. Historical Simulation with Full Valuation is a more complex engine required for non-linear derivatives, like options.

SUBADDITTIVITY

Subadditivity states that a portfolio's total risk must never exceed the sum of its individual parts. This crucial diversification rule is mathematically guaranteed in Parametric (Sigma Flat/EWMA) and most Monte Carlo models. However, Historical VaR is the only standard method that can actually violate this fundamental property (example: portfolio with Apple and Amazon). If Asset A and Asset B experience extreme crashes on *different* days, combining them increases the overall frequency of disasters in the portfolio. This can push these massive losses past the VaR percentile threshold, causing the combined portfolio to mathematically appear riskier than its isolated parts. In our case it does not happen as IBIT and LVMH have negative correlation, meaning their worst historical days rarely overlap,

CONCLUSION – BENEFITS OF DIVERSIFICATION

Since all our calculated VaRs demonstrate subadditivity, the mathematical benefit of diversification is fully validated for our portfolio. This is especially powerful given that IBIT and LVMH share a negative correlation, meaning their price movements naturally offset each other during market shocks. Consequently, combining these assets actively reduces our overall risk exposure, proving that diversifying is a highly effective strategy.

Although we notice that the IBIT results demonstrate that a Gaussian distribution fails to capture the asset's actual risk profile due to its excessively fat tails.

